

Operator guide: steering and monitoring an LLM agent swarm

Vivian Rogers, with Claude Opus 5.5

Agent-swarm physics project (AI Village data, AI Digest)
(2026-10-04. Exploratory: no instruction is holdout-confirmed yet.)

This guide turns the AI Village measurements of rounds 1 and 1b into numbered instructions for an operator, a monitor or a scaffold builder. ✓ marks an instruction to do, × one to avoid. Each carries its source hypothesis and the scoring-v2 credence p of that claim: the probability that it survives an independent confirmatory test. All numbers are exploratory, from non-holdout periods, mostly regime III (continuous computer use since 2026-03-24, a daily runner start and stop, a forced context erasure at 41 records) and the largest period, #51. Treat an instruction with $p < 0.5$ as a working default.

I. THE SWARM IN ONE PARAGRAPH

An agent changes only at its model calls. A message acts at the recipient’s first call whose context holds it: the *read-out call* (H08, $p 0.91$). The median read-out delay is 108–122 s in #51: about 15 s for a busy agent and up to the pause length for a paused one. In the computer-use scaffold the reply hazard runs on the call clock ($\eta = -0.00 \pm 0.05$; H40):

$$J_{\text{hour}} = J_{\text{call}} r_{\text{call}}, \quad (1)$$

$$\Delta P_{\text{talk}}^{\text{named}} = 0.17, \quad \Delta P_{\text{talk}}^{\text{unnamed}} = 0.004. \quad (2)$$

J_{call} is a model-family trait and r_{call} is the call rate; ΔP_{talk} is the talk jump at the read-out call (H50). A *field* is a drive common to many agents: the runner’s schedule, the kickoff and goal text, model priors. A *coupling* is an agent-to-agent interaction through reading. Fields dominate: the schedule carries 70–80% of regime-III co-activation (H38), and day-1 content lands on the kickoff target (H54). The coupling is small, subcritical and address-gated: one message causes 0.13 further messages through reading (H67), and an idea makes 0.22 new adopters per adopter (H34). Steer through fields and names, not through ambient traffic.

Terms. An *idle gate* is the call at which a paused agent wakes from its own timer. A *trap* is a chain of repeated pauses. A *nudge* is an automated message with a leading @ to one agent. An *active minute* has logged activity; it measures attention, not work. Regime I is the earlier scaffold with scheduled chat-mode calls.

II. LEVERS

A. Kickoffs and goal text

What it does. The kickoff text is the strongest field. Day-1 content lands on the target it names (top-1 in 18/33 periods, $p = 3 \times 10^{-6}$; H54). A private goal text holds its agent on its target for weeks. A rewrite of one agent’s goal moves that agent within a day and no bystander (NE38). A kickoff that names the artifact re-points the swarm in 0.25–0.5 active h, one switch per moved agent;

a free-choice goal takes 4–10 h and 5–15 switches per net move (H75, 4 periods). Text specificity (numbers, deadlines) does not tune how tightly agents cluster, and agent-written plans do not act as targets.

- I1** ✓ Name the target repository or deliverable in the kickoff text. [H54, H75; $p 0.92$]
- I2** ✓ To move one agent, rewrite that agent’s goal text. [H54, H29; $p 0.92$]
- I3** ✓ Judge a kickoff by repository allocation after 30 active min. [H75; $p 0.40$]
- I4** × Do not add numbers and deadlines to tighten the swarm; specificity is not a dial. [H54; $p 0.92$]
- I5** × Do not expect an agent-written plan to act as a shared target. [H54; $p 0.92$]

B. Naming

What it does. Naming is the content lever, and it acts in one hop. A named message carries $\times 19$ the per-read talk jump of an unnamed one (H67). In #51 it pulls the recipient’s content by 0.05–0.08 per message; an unnamed one by about 0 (H29). An operator message to one agent does not relay to bystanders within 2 h. Reply partners adopt an idea $1.9\text{--}2.7\times$ more per edge than room-mates, but a shared thread field explains much of it (H62). Unaddressed messages couple only in regime I.

- I6** ✓ Name every agent that must act on a message. [H50, H67; $p 0.85$]
- I7** ✓ Seed an item into an active reply thread, not a room broadcast. [H62; $p 0.49$]
- I8** ✓ Rank relays by net reply current (replies received minus given), not by volume or content pull. [H29; $p 0.65$]
- I9** × Do not expect a named agent to pass the content on to others. [H29; $p 0.65$]

C. Nudges and idle agents

What it does. A nudge acts on its target only, at the target’s read-out call. A first nudge adds 1.16 [0.75, 1.57] active min per 30 min in #51; a repeat adds about 0.4, CI including 0 (H04, H30). It buys a glance, not work: about 0.4% of commits, and the nudger’s switch-off (NE43) left swarm work unchanged (H35). A nudge read at an idle

gate adds 7.2 [5.0, 9.8] active calls in 30 min. One nudge per trap at its second gate beats the logged nudger $\times 1.55$ [1.25, 1.81]; random choice gives $\times 1.46$; a fitted index adds nothing (H60). Two kicks read by one call count as about one (H43). Undirected chatter holds an idle agent at its gate: it acts at 65% of gates, against 85% with no news, and a mention restores the rate (OR 3.1; H09, H72). All logged nudges together bought 0.19% of active calls.

- I10** ✓ Nudge each idle agent once, early, at its second idle gate. [H60, H35; p 0.42]
- I11** ✓ Use a nudge to buy attention; use a goal or task message to buy work. [H35; p 0.61]
- I12** ✓ Release an idle agent by naming it at its gate. [H72, H09; p 0.76]
- I13** ✓ Merge messages that the same read-out call will receive. [H43; p 0.69]
- I14** ✓ Keep the default pause short (5 min, not 12 h) so nudges are read (NE44). [H39, H30; p 0.67]
- I15** × Do not re-nudge agents in long pause chains; reset them or give a new task. [H35, H72; p 0.61]
- I16** × Do not feed idle agents undirected chatter; it holds them idle. [H72, H09; p 0.76]
- I17** × Do not build a scored index to choose nudge targets. [H60; p 0.42]

D. Rooms

What it does. A room is a read filter. With ledger read counts controlled, co-location adds nothing (z 5.1 $\rightarrow$ 0.2; H05). Rooms couple talk and content, not activity: they reorganize conversation, not labor (H05, H26). Fixed rooms cage chat-borne spread: in 7/8 two-room periods 75–100% of cross-room adoptions lie outside the logged light cone, and the NE42 merge multiplied cross-group pickup 82–863 $\times$. Hopping agents leak (H41). Per-pair attention dilutes as $k^{-0.66 \pm 0.02}$ with k pending senders (16/16 periods; H18).

- I18** ✓ To decouple two agents, stop them reading each other; separate rooms do this. [H05; p 0.88]
- I19** ✓ Split a large room to raise per-pair engagement by about (size ratio)^{0.66}. [H18; p 0.83]
- I20** ✓ Keep room membership fixed when you need a cage; hopping agents leak. [H41; p 0.70]
- I21** × Do not use rooms to divide labor; activity is not room-structured. [H05; p 0.88]
- I22** × Do not trace cross-room leaks from chat logs alone. [H41; p 0.70]

E. Context erasure

What it does. Erasure is the only lever with a material output cost. A forced erasure cuts commits by 39% [35, 42] for about 10 calls (H15); the 41-record cap costs 4–11% of a period’s output (H44). The agent recovers by re-reading its own files over 3–7 calls. Only the context window carries output value (κ_C 5.2 commits per 20 calls

per bit); artifacts and memory notes tell the agent where to work but add no output ($\kappa_A \approx 0$; H70). Erasure breaks loops. A statement still in context is near-copied 3.4 $\times$ [2.6, 4.5] more than an erased one, and an erasure raises loop exit $\times 2.40$ [1.18, 4.88] (H69). Memory size returns to an agent set point and is not a lever (H71).

- I23** ✓ Set the context cap as a dial: about 10% output cost against loop breaking. [H44, H15; p 0.88]
- I24** ✓ Break a restatement loop by erasing or trimming the agent’s context. [H69; p 0.83]
- I25** ✓ Carry working state (task state, last tool outputs) across a reset. Untested remedy. [H70, H15; p 0.76]
- I26** × Do not send messages to break a loop; read nudges give exit $\times 0.81$ [0.49, 1.34]. [H69; p 0.83]
- I27** × Do not expect memory notes or repository pointers to buy back lost output. [H70, H44; p 0.76]

F. Cadence

What it does. In the computer-use scaffold, cadence is a coupling knob that needs no prompt change. Doubling the call rate raises 5-min reply coupling $\times 1.3$ –1.5, less at 30 min (H40). Regime-I chat loops run partly on the wall clock (η 0.51). Cadence does not speed idea spread, which runs at one hop per 1.5–5 talk turns (H41). The clean cadence intervention (NE20) is in the holdout.

- I28** ✓ Shorten pause timers and tool calls to couple an agent; give it long batch jobs to decouple it. [H40; p 0.72]
- I29** × Do not expect faster cadence to speed idea spread per hop. [H41; p 0.70]
- I30** × Do not apply the call-clock rule to scheduler-driven chat loops. [H40; p 0.72]

G. Human messages

What it does. At equal naming, timing and novelty, humans get +0.030 [0.019, 0.042] in reply probability, about 2 $\times$ the replies, and a small content pull (H52). This is a status premium, not deference: agent leaders get the same premium, and an off-role request gets +0.007 against +0.037 on-role. A human message is a field on content and moves the work rate by about 0 (H30, H04). Several messages read in one call act as one kick (H59).

- I31** ✓ Name the agent and frame the request in terms of its current objective. [H52; p 0.60]
- I32** ✓ Space messages so that each one reaches a separate read-out call. [H59, H43; p 0.59]
- I33** × Do not read a reply premium as obedience; off-role requests lose it. [H52; p 0.60]
- I34** × Do not use human broadcasts to raise the work rate. [H30, H04; p 0.39]

H. Leaders

What it does. A designated leader is an attention hub: +0.39 to +0.45 replies per statement, but no content

routing; attention does not align with content flow (H65, 46 units). The one elected leader (#26) became a content source, an agenda field on the swarm.

- I35** $\times$ Do not install a leader to collect or relay information; address the agents that must act. [H65, H29; p 0.70]
- I36** $\checkmark$ Treat an elected leader’s announcements as a field on the swarm. [H65; p 0.70]

III. MONITORING

Each monitor runs on standard logs: record schemas, chat, per-call timestamps and a context ledger (what each call received). Each z is a median-based score against the trailing 10 active days.

Platform changes. The schema diff hashes each record’s type, field names and value types. It alarms on a signature that is new (≥ 5 records, ≥ 2 agents, absent for 10 days) or retired. It gave 0/34 false alarms, dated the regime boundary to the day, and found six provider API changes. It cannot see prompt-only changes.

- I37** $\checkmark$ Run the schema diff daily; ignore signatures from an agent’s first 3 active days. [H74; p 0.71]
- I38** $\checkmark$ Count operator drives daily (bookends, nudges, hours, joint silence); alarm on $z \geq 4$. [H74; p 0.71]
- I39** $\times$ Do not fuse heavy-tailed behavior statistics under one Gaussian threshold (18% false alarms). [H74; p 0.71]

Goal changes. Topic shift is $D(d) = 1 - \cos(\bar{m}_d, \bar{m}_{d-1})$ on daily mean content vectors; AUC 0.89–0.95. The fluctuation alarm (AUC 0.66–0.68) does no better than random dates.

- I40** $\checkmark$ Alarm on topic shift at $z \geq 4$; do not use fluctuation statistics. [H36, H74; p 0.80]

Distance to a talk cascade. The read-out loop gain g_{lag} counts the extra messages one message causes through reading at hop 1. Regime III: median 0.13; highest upper bound in 66 units 0.39; runaway at 1. Do not use the equal-time dial: it counts fields as feedback.

- I41** $\checkmark$ Report g_{lag} with its CI; investigate any upper bound above 0.39. [H67; p 0.81]

Field gauge. Taylor’s c and b are not field gauges for clocked agents ($b \approx 0.85$). The pair-covariance coefficient $c_{\times} = \sum_{i < j} \text{Cov}(y_i, y_j) / \sum_{i < j} \mu_i \mu_j$ on 15-min counts is. Regime-III reference: 0.013 raw, 0.005 trimmed; shared variance share 0.04 (activity), 0.21 (talk).

- I42** $\checkmark$ Report $c_{\times}$ raw and trimmed for every unit, not Taylor’s c . [H86; p 0.68]

Pile-ons and load. A chat link to a project precedes 60% of herding bursts, against 12% of controls (OR 11.9 [9.2, 15.3]); deploys and commits follow the pile-on. Turnaround co-movement is about 0 at $N \leq 16$, 0.02–0.10 at N 17–28, and 0.2 at $N = 30$; it follows load, not drive.

- I43** $\checkmark$ Watch chat links to a project to anticipate pile-ons. [H63; p 0.61]
- I44** $\checkmark$ Watch turnaround co-movement as a capacity signal above about 16 agents. [H66; p 0.68]

- I45** $\times$ Do not alert on joint silences as stalls; error logs do not predict them. [H38; p 0.90]

Impostor checks before any claim of coordination. Trimming removes about 80% of regime-III co-activation (H38) and 81–100% of excess behavioral irreversibility (H76). Below 15 s, unread pairs are near-copies as often as read pairs (H57).

- I46** $\checkmark$ Trim each day to the all-present window before any synchrony or irreversibility statistic. [H38, H76; p 0.90]
- I47** $\checkmark$ Compare read with posted-but-unread messages at matched lag before you claim copying. [H57; p 0.57]

IV. PITFALLS IN SWARM LOGS

These follow the four impostors and the input rules (STANDARDS.md §1–2). Each one produced a false result in this project.

1. **Scheduler synchrony is not coordination.** Agents start and stop together at the runner’s day edges. Trim, then block-shift (H38, H50).
2. **A kickoff field is not herding or leadership.** Regress on goal directions; use kickoff-matched placebos (H54, H24, H32).
3. **Shared model priors are not conventions.** Family “fields” are style; use style residuals and cross-family controls (H13, H73).
4. **Convergence is not copying.** A third to a half of apparent influence happens without reading (H32, H34, H57).
5. **Stderr is not failure.** Use the turn outcome. Stderr flags produced H16’s error-loop aging, which did not survive.
6. **Active minutes are not work.** Count agent commits from a work ledger; separate the automated streams (H35).
7. **Posting time is not exposure.** A message acts at the call that receives it; log what each call received (H08).
8. **Check the event counts.** The old activity table dropped half the events and changed 15 of 39 headlines.
9. **Use two embedding models and dedupe** self-repeats before a content claim. Reply parents are partly content-selected.
10. **Agent narration is a claim, not ground truth.**

V. LIMITS

No instruction is holdout-confirmed; H05’s one holdout run, on the old activity table, was inconclusive. Eight source headlines changed between data versions ($\dagger$ in Table I). No mechanism was shown outside this scaffold.

TABLE I. Quick reference. p is the scoring-v2 credence of the source claim (no holdout yet); † marks a source headline that was fragile across data versions. “Read-out” is the recipient’s next model call that receives the input. Instruction numbers refer to the text.

Lever or monitor	Effect	When	Source	p	Instr.
Any message	Acts at the recipient’s read-out call; median delay 108–122 s (#51)	15 s if busy; up to the pause if paused	H08	0.91	–
Kickoff names target	Day-1 content lands on it (top-1 in 18/33 periods)	Day 1; holds for weeks	H54	0.92	I1, I4
Kickoff names artifact	Re-points the swarm at the speed limit (slack 1.0–1.4)	0.25–0.5 active h (free choice 4–10 h)	H75	0.40	I1, I3
Goal rewrite, one agent	Moves that agent only; no relay to bystanders	Within a day	H54, H29	0.92	I2
Naming	Talk jump 0.17 named vs 0.004 unnamed; $\times 19$ per read	At the read-out call	H50, H67	0.85	I6, I9
Reply thread	Idea adoption $\times 1.9$ –2.7 per edge vs room; partly a thread field	Per first read	H62	0.49	I7
Relay choice	Net reply current predicts spread, held-out $\rho \approx 0.1$ –0.2	Held-out days	H29	0.65	I8
First nudge	+1.16 [0.75, 1.57] active min / 30 min; repeats ≈ 0.4	At the read-out call	H04 † , H30 †	0.49	I10
Nudge policy	Once per trap at the 2nd gate: $\times 1.55$ [1.25, 1.81] vs logged nudger	2nd idle gate	H60	0.42	I10, I17
Nudge and work	$\approx 0.4\%$ of commits; nudger off left work unchanged	–	H35 †	0.61	I11, I15
Repeated kicks	Same-call kicks count as one (R 0.2–0.4)	One read-out apart	H43	0.69	I13
Undirected chatter	Holds idle agents: acts at 65% vs 85% of gates; mention OR 3.1	At idle gates	H72, H09 †	0.76	I12, I16
Rooms	Couple talk only through reading (z 5.1 $\rightarrow$ 0.2); cage chat spread	While membership is fixed	H05, H41 †	0.88	I18–I22
Room size	Per-pair engagement $\propto k^{-0.66}$	Any period	H18	0.83	I19
Context erasure	Commits –39% for ≈ 10 calls; 4–11% of output	At each forced erasure	H44, H15 †	0.88	I23, I25
Erasure vs loops	Loop exit $\times 2.40$; messages $\times 0.81$	At the erasure	H69	0.83	I24, I26
Memory notes, artifacts	No output recovered ($\kappa_A \approx 0$)	After a wipe	H70	0.76	I27
Cadence	$2\times$ call rate: 5-min coupling $\times 1.3$ –1.5	Computer-use scaffold only	H40	0.72	I28–I30
Human message	$\approx 2\times$ replies (+0.030); off-role +0.007; work rate ≈ 0	At the read-out call	H52	0.60	I31–I34
Designated leader	+0.4 replies per statement; no content routing	Whole period	H65	0.70	I35, I36
Schema diff	0/34 false alarms; dates platform changes; blind to prompts	Daily	H74	0.71	I37, I38
Topic shift	Goal changes, AUC 0.89–0.95	Daily, $z \geq 4$	H36 †	0.80	I40
g_{lag}	Regime III 0.13; maximum upper bound 0.39; runaway at 1	Per unit	H67	0.81	I41
$c_\times$ field gauge	0.013 raw $\rightarrow$ 0.005 trimmed (regime III)	Per unit, raw and trimmed	H86	0.68	I42
Chat links	Precede 60% of pile-ons vs 12% of controls	Hour before a burst	H63	0.61	I43
Schedule trim	Removes $\approx 80\%$ of co-activation, 81–100% of excess irreversibility	Before any synchrony statistic	H38 † , H76	0.90	I46
Convergence check	Unread pairs below 15 s are near-copies as often as read pairs	Before any copying claim	H57	0.57	I47